

Santhosh Narayanan Baburaman

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EDUCATION

Master of Science in Analytics, University of Southern California (GPA: 3.92/4) Jan 2025 – Present
Bachelor of Technology in Computer Science and Engineering, VIT (CGPA: 8.41/10) Sep 2020 – Apr 2024

EXPERIENCE

Founder & President – ISE Analytics Student Association (IASA) Feb 2025 – Present

- Founded and led the ISE Analytics Student Association at USC through the university RSO recognition process
- Organized 12+ professional events (alumni panels, workshops, networking) engaging 500+ students
- Coordinated 20+ industry professionals and alumni to deliver career insights and networking opportunities

AI Intern – SmartInternz May 2023 – Jul 2023

- Built an end-to-end CNN-LSTM image captioning workflow on the Flickr8k dataset using structured preprocessing and hyperparameter tuning
- Developed a Flask-based prototype that generated captions for uploaded images, enabling interactive testing of model outputs
- Led a 3-member team coordinating model pipeline design, training iterations, and error analysis

PROJECTS

Automated Sales Analytics Copilot (Python, PostgreSQL, Claude API, Streamlit)

- Designed a PostgreSQL star schema with 5 KPI views tracking revenue, margin, and discount impact across product categories and US regions on a 10K-order, \$2.3M dataset
- Automated anomaly detection and forecasting across 4 years of sales data, surfaced that Furniture margins collapse to -22% under high discounts; alerts delivered nightly via Slack/email
- Built a natural language chat UI (Claude AI + Streamlit) enabling business users to query KPIs, run discount what-if scenarios, and generate ad-hoc SQL reports without writing code

Financial Analytics & Power BI Dashboard (Python, PostgreSQL, Power BI)

- Built an end-to-end ETL pipeline extracting 10-K/10-Q financial data via the SEC EDGAR API and transforming filings using Python (Pandas)
- Modeled cleaned financial data in a PostgreSQL star schema and wrote SQL queries for aggregation and trend analysis
- Built an interactive Power BI dashboard visualizing multi-company revenue, profit, and asset trends (2010–2025)

Sales Performance & Profitability Analysis Dashboard (Tableau)

- Designed an interactive Tableau dashboard analyzing \$12M+ in sales across regions and product categories
- Built KPI dashboards tracking revenue, profit margin, and order volume with dynamic filters and drill-down analysis
- Identified key profitability drivers, uncovering negative margins in specific categories and losses from high discounting

NYC Taxi Demand & Fare Analysis (Python, Machine Learning, Time Series)

- Analyzed 1M+ NYC taxi trip records to model travel time, fare, and traffic speed using machine learning and time series approaches
- Engineered features and performed correlation analysis to identify key trip and temporal variables driving urban mobility patterns
- Co-authored an IEEE-published paper on predictive analytics for transportation and traffic forecasting

SKILLS

- **Data & Analytics:** Python, SQL, PostgreSQL, Snowflake, BigQuery, ETL, Data Modeling, Pandas, R
- **Visualization & Reporting:** Power BI, Tableau, Streamlit, Microsoft Excel
- **Analytics Methods:** Predictive Modeling, Feature Engineering, Time Series Analysis, Anomaly Detection
- **ML & Tools:** NLP, DistilBERT, ARIMA, Prophet, LLM APIs, GCP (Vertex AI, AutoML), Azure, Git